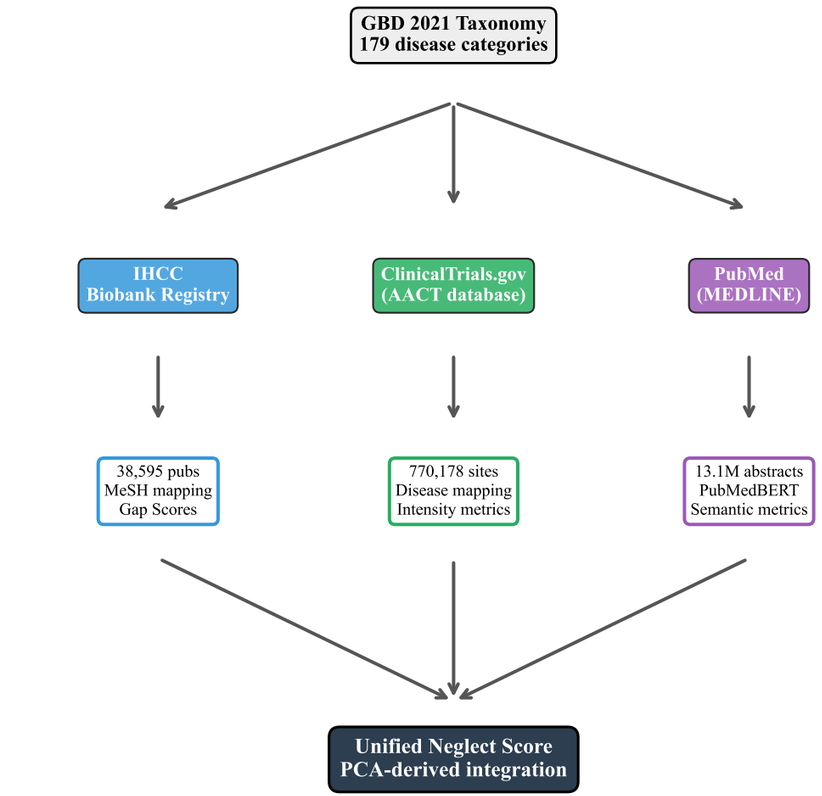
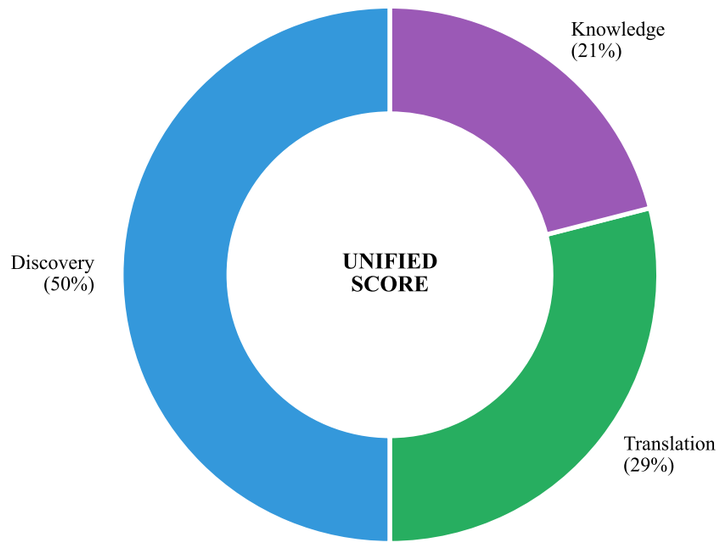
a

Discovery	Translation	Knowledge
Biobanks	*Clinical Trials*	*Scientific Literature*
70 biobanks	**563,725 trials**	**13.1 million abstracts**
29 countries	**770,178 sites**	**175 diseases**
38,595 publications	**194 countries**	**PubMedBERT embeddings**
Key metrics:	**Key metrics:**	**Key metrics:**
Gap Score: research coverage vs. disease burden	**Research Intensity:** trials per million DALYs	**Semantic Isolation Index (SII):** distance from other disease research
Equity Alignment Score (EAS): biobank-level equity	**Geographic Ratio:** HIC vs. LMIC trial site distribution	**Knowledge Transfer Potential (KTP):** cross-disease similarity

b



c



PC1 explains 63.3% of variance
n = 86 diseases, 3 dimensions